02 - Image Preprocessing Pipeline

Milestone 2: Preparing the Rakuten product images for reproducible image and multimodal modelling.

This notebook converts the image-audit findings into a reusable preprocessing workflow. It does not train an image model. Raw files remain unchanged, and images are opened only when they are needed.

1. Scope and Audit Context

The data-audit notebooks established the following points:

- all 84,916 labelled training products have a matching image file;
- all 2,000 sampled images were readable, RGB images with a size of 500 × 500 pixels;
- exact file duplicates were audited for the labelled training catalogue only;
- the portable training-image file hashes were cached under `data/processed/`.

The purpose of this notebook is therefore not to repeat the image audit. It prepares the image manifest, loads the shared validation split created by `02_preprocessing.ipynb`, uses the cached audit metadata to create a cleaned image training set while keeping validation unchanged, and defines deterministic framework-neutral image preprocessing.

2. Imports and Configuration

Project paths and image filenames remain centralised in `src.data_utils`. No dataset path is duplicated in this notebook.

```
Environment and data utilities loaded successfully.
```

3. Build the Training Image Manifest

`load_merged_data()` returns the labelled challenge data. The original dataframe index is retained as `row_id`; this is the safest key for joining text predictions, image predictions and split assignments later in the project.

The manifest stores metadata and file paths only. It does not load 84,916 images into memory.

```
Training image rows: 84916
```

	row_id	imageid	productid	prdtypecode	source_split	image_path
0	0	1263597046	3804725264	10	train	/Users/wahajaslam/capstone_rakuten/data/raw/i...
1	1	1008141237	436067568	2280	train	/Users/wahajaslam/capstone_rakuten/data/raw/i...
2	2	938777978	201115110	50	train	/Users/wahajaslam/capstone_rakuten/data/raw/i...
3	3	457047496	50418756	1280	train	/Users/wahajaslam/capstone_rakuten/data/raw/i...
4	4	1077757786	278535884	2705	train	/Users/wahajaslam/capstone_rakuten/data/raw/i...

4. Validate the Manifest

The earlier audit already checked the complete training filename coverage. These assertions protect the preprocessing workflow against accidental path, join or data changes.

```
Available training images: 84916/84916
Missing training images  : 0
```

5. Load the Shared Training/Validation Split

The official challenge test data has no labels and is not used for local evaluation. This notebook therefore loads the shared training/validation assignment created from the labelled catalogue in `02_preprocessing.ipynb`. The assignment remains unchanged here so text and image predictions can still be joined through `row_id`.

split	products	classes	share
train	67932	27	80.0%
validation	16984	27	20.0%

6. Create Duplicate- and Conflict-Free Image Sets

The cached file SHA-256 hashes identify exact image matches. The shared validation set remains completely unchanged so that text and image models are evaluated on exactly the same products. Cleaning is restricted to training and does not overwrite the shared split. The following deterministic rules are applied:

1. keep every row assigned to validation;
2. remove training rows whose file hash also occurs in validation to prevent leakage;
3. remove conflicting file-hash groups from training only;
4. keep one training row per remaining file hash and retain `row_id` for multimodal comparison.

The raw data and the shared `data_split.csv` remain unchanged.

	before	after	classes	unique_file_hashes	removed	removed_share
train	67932	62312	27	62312	5620	8.27%
validation	16984	16984	27	16603	0	0.00%

prdtypecode	train_count	validation_count	train_share	validation_share	absolute_share_difference
10	2483	623	3.98%	3.67%	0.32%
40	1978	502	3.17%	2.96%	0.22%
50	1272	336	2.04%	1.98%	0.06%
60	623	166	1.00%	0.98%	0.02%
1140	2127	534	3.41%	3.14%	0.27%
1160	3135	791	5.03%	4.66%	0.37%
1180	605	153	0.97%	0.90%	0.07%
1280	3528	974	5.66%	5.73%	0.07%
1281	1601	414	2.57%	2.44%	0.13%
1300	3110	1009	4.99%	5.94%	0.95%
1301	590	161	0.95%	0.95%	0.00%
1302	1928	498	3.09%	2.93%	0.16%
1320	2556	648	4.10%	3.82%	0.29%
1560	3659	1015	5.87%	5.98%	0.10%
1920	3304	861	5.30%	5.07%	0.23%
1940	632	161	1.01%	0.95%	0.07%
2060	3836	999	6.16%	5.88%	0.27%
2220	643	165	1.03%	0.97%	0.06%
2280	3808	952	6.11%	5.61%	0.51%
2403	3817	955	6.13%	5.62%	0.50%
2462	1133	284	1.82%	1.67%	0.15%
2522	3667	998	5.88%	5.88%	0.01%
2582	1895	518	3.04%	3.05%	0.01%
2583	5783	2042	9.28%	12.02%	2.74%
2585	1699	499	2.73%	2.94%	0.21%
2705	2208	552	3.54%	3.25%	0.29%
2905	692	174	1.11%	1.02%	0.09%

```
Excluded conflict rows: 676
Excluded rows in total: 5620
```

7. Export the Clean Image Split

The model-facing manifest contains identifiers, labels, split assignments and file hashes only. Absolute image paths are not stored, so each machine can reconstruct them with `get_image_path()`.

```
Rows: 79296
```